course-project-2

- Title : Smart Door unlocking system Using Verilog HDL
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Abstract:

The Smart Door Unlocking System is designed to provide secure and automatic access control using a password-based sequence detection method. The system uses a Finite State Machine (FSM) implemented in Verilog to verify the correct input pattern in real time. When the correct sequence is entered, the system generates an unlock signal, ensuring reliable and efficient operation. This design improves security, reduces manual effort, and can be integrated into smart home and IoT applications.

Introduction

The Smart Door Unlocking System is developed to provide a secure, automated, and convenient method of access control using digital technology. Traditional locks often face problems such as key loss, duplication, and limited security, which modern environments can no longer rely on. By using a predefined input sequence and implementing it through a Finite State Machine (FSM), this system ensures accurate pattern detection and reliable unlocking. It offers improved safety, faster operation, and easy integration with smart home or IoT platforms, making it a modern solution for efficient and secure access management.

Background

The shift from traditional mechanical locks to digital security solutions has increased due to rising concerns about safety, convenience, and reliability. Mechanical locks are prone to issues like key loss, duplication, and physical wear, making them less effective for modern security needs. With advances in digital electronics, sequence detectors and FSM-based authentication systems have become popular because they offer accurate, fast, and secure access control. These digital methods are easy to implement on hardware platforms like FPGA or microcontrollers, making them ideal for smart homes and IoT applications. This project builds on these developments to create a secure and efficient Smart Door Unlocking System.

Why Smart unlocking?

Smart unlocking is important because it offers higher security, convenience, and automation compared to traditional mechanical locks. Physical keys can be lost, copied, or damaged, but smart systems use digital authentication such as passwords or biometric patterns, making unauthorized access more difficult. They also provide fast, keyless entry and can integrate with IoT devices for remote control and monitoring. Overall, smart unlocking gives a more reliable, modern, and efficient solution for securing homes, offices, and smart environments.

Purpose of the Project

The purpose of this project is to design a secure and efficient Smart Door Unlocking System that replaces traditional mechanical locks with a digital, sequence-based authentication method. By using an FSM to verify the correct input pattern, the system ensures accurate unlocking, minimizes unauthorized access, and enhances overall security. This project aims to provide a reliable, low-power, and easy-to-implement solution suitable for smart homes, offices, and IoT-based security applications.

Objective

The main objective of this project is to design and implement a secure Smart Door Unlocking System that accurately detects a predefined input sequence using an FSM-based approach. The system aims to enhance safety by preventing unauthorized access, provide a reliable digital alternative to mechanical locks, and ensure fast, error-free unlocking. It also focuses on creating a low-power, cost-effective, and easily integrable solution suitable for smart homes, offices, and IoT-based security applications.

Verilog Code Description

Code :

```
module door_unlock(
input clk,reset,x,
output reg z
 );
 parameter s0=3'd0,s1=3'd1,s2=3'd2,s3=3'd3,s4=3'd4;
 reg[3:0] state, next_state;
 always @(posedge clk or posedge reset)
 begin
 if(reset)
 state <= s0;
 else state<=next_state;
 end
 always @(*) begin
 case(state)
 s0:next_state =(x)?s1:s0;
 s1:next_state =(x)?s1:s2;
 s2:next_state =(x)?s3:s0;
```

```
 s3:next_state =(x)?s4:s2;
 s4:next_state =(x)?s1:s2;
 default:next_state=s0;
 endcase
 end
 always @(*) begin
 case(state)
 s4:z=1'b1;
 default:z=1'b0;
 endcase
 end
endmodule
```

Module explanation

The door_unlock module is a **Moore Finite State Machine (FSM)** designed to detect a specific input sequence and produce an unlock signal (z = 1) only when the correct pattern is received. The module uses five states (s0 to s4), where s0 represents the initial/reset state, and s4 is the final unlock state.

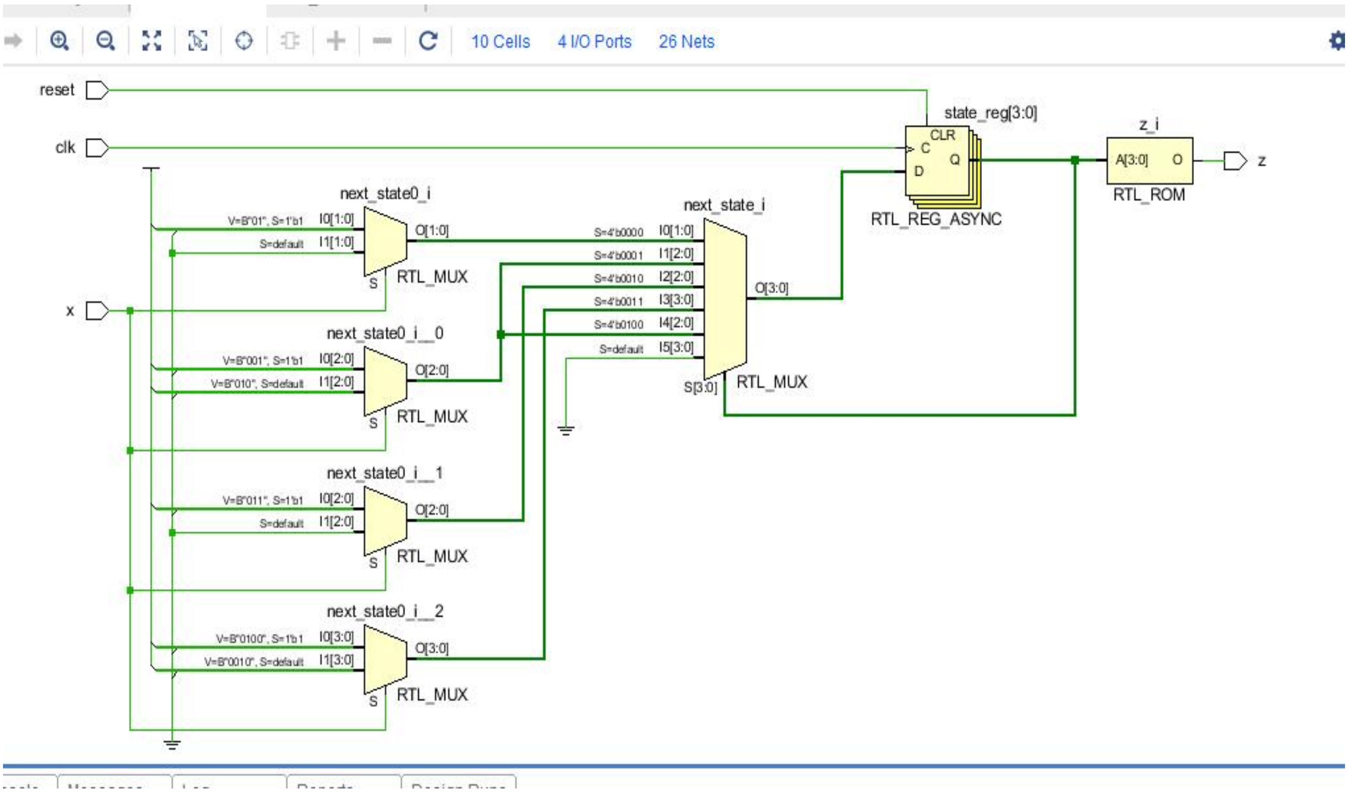
The FSM has three main parts:

1.State Register (Sequential Block):
This block updates the current state on every positive edge of the clock. When the reset signal is high, the system returns to the initial state s0. Otherwise, the FSM moves to next_state.

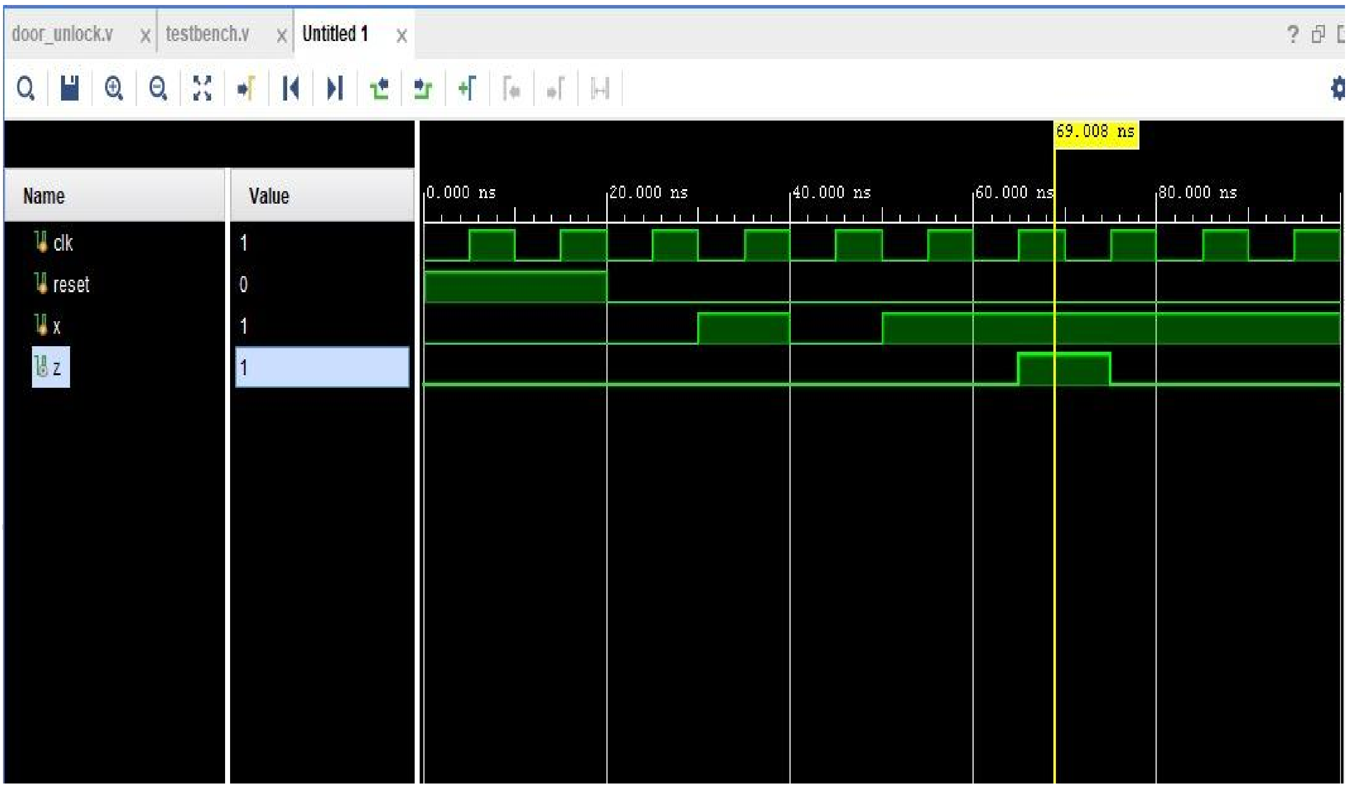
2.Next-State Logic (Combinational Block):
This block determines how the FSM transitions between states based on the input x. Each state checks whether the input is 0 or 1 and moves to the appropriate next state. These transitions represent the sequence detection logic of the system.

3.Output Logic (Combinational Block):
Since it is a Moore machine, the output depends only on the current state. The output z becomes 1 only when the FSM reaches state s4, indicating that the correct sequence has been detected. For all other states, z = 0.

SCHEMATIC DIAGRAM:

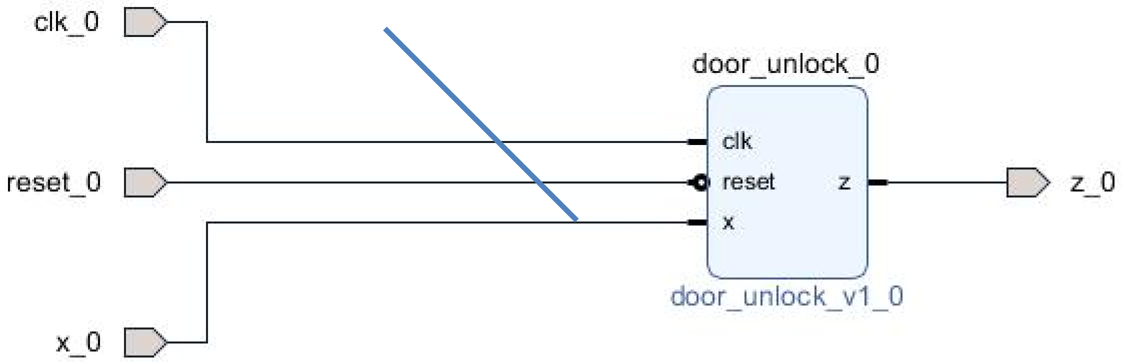


SIMULATION WAVEFORM:



IP (Intellectual Property):

IP (Intellectual Property) integration is the process of embedding reusable design blocks into a larger digital system. In this project, the Smart Door Unlocking System is designed as an **FSM-based sequence detector**, which can be packaged and integrated as a reusable **Verilog IP core** in SoC/FPGA platforms.



Conclusion:

The Smart Door Unlocking System successfully demonstrates how an FSM-based sequence detector can enhance security through a reliable and automated unlocking mechanism. By using digital logic and state transitions, the system ensures that the door unlocks only when the correct input pattern is received, reducing the chances of unauthorized access. Its low-cost design, easy implementation, and ability to be integrated as reusable IP make it suitable for modern smart home and IoT applications. Overall, the project provides an efficient, secure, and scalable solution for smart access control systems.